

Project 1a

NUMN21/FMNN25: Advanced Numerical Algorithms in Python
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This describes an add-on to Project 1. We expect that Project 1 has already progressed. This add-on is intended to check the extendibility of your code, i.e. can the present code be extended to also solve for a different layout of rooms or are there parts of the code, which have to be rewritten?

This assignment has 2 tasks.

Background

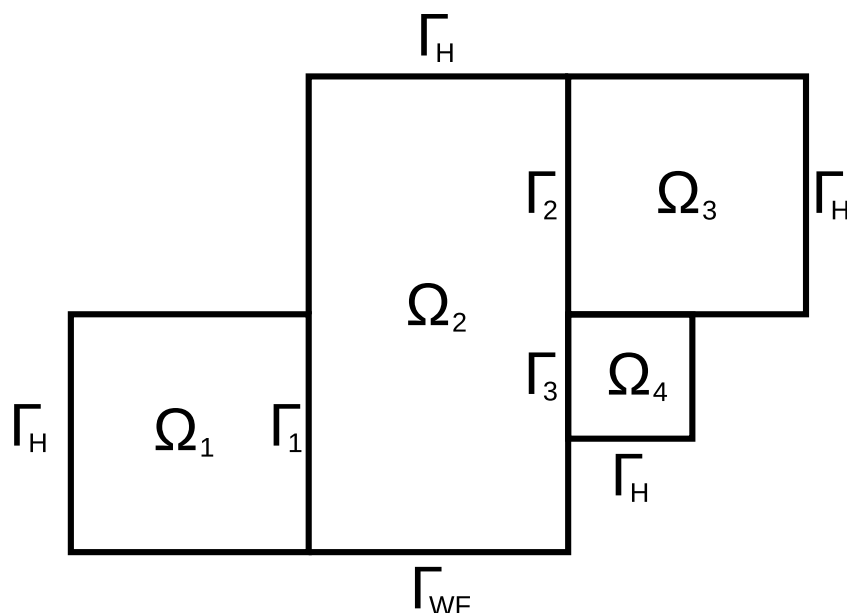


Figure 1: Layout of the 2,5-room apartment. Rooms 1 and 3 are of size 1×1 , room 2 is of size 1×2 and room 4 is $1/4$ the size of room 1 and 3

Task 1

During the bidding process for the 2-room apartment you were unsuccessful and somebody else bought it. A new apartment with a very similar layout has just come on the market. Repeat the exercise from Project 1 with the new layout.

Task 2

Try to compute the solution using $\Delta x = \frac{1}{100}$.